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Chapter 1

Introduction

1.1 Context - The Need for Parametric Solutions

Systemically important agricultural sectors remain exposed to unmanageable, non-diversifiable weather risk that cannot be hedged with standard financial instruments. Indian rice farming is a paradigmatic case: a vast production system essential to food security and rural livelihoods, yet tightly coupled to the volatility of the monsoon. Shocks in the timing, intensity, or spatial distribution of rainfall propagate directly into yield and income variability. Because these shocks are co-variate, many producers are hit simultaneously—the resulting losses are macroscopic and often outside the control of individual farmers.

Traditional indemnity-based insurance has repeatedly failed to address this exposure due to three well-documented frictions. First, **moral hazard**: when payouts depend on realized on-farm losses, full coverage can dampen incentives to incur costly mitigation (e.g., emergency irrigation) during stress. Second, **adverse selection**: producers with private information about higher local risk are more likely to buy coverage, worsening the insurer's pool. Third, **high transaction costs**: verifying plot-level losses across millions of small, geographically dispersed farms is slow, expensive, and prone to dispute—delaying liquidity precisely when it is most needed.

Parametric (index-based) derivatives target these frictions by tying payouts to an objective, transparent, and publicly verifiable index—specifically, cumulative rainfall derived from gridded satellite data. Because farmers cannot influence rainfall, moral hazard collapses; because the trigger is uniform and observable, adverse selection is muted; and because no field-level loss assessment is required, claims adjustment costs are minimal. This thesis explores the viability of such an instrument in the Kanker district of Chhattisgarh.

1.2 The Problem

While parametric derivatives theoretically solve the issues of moral hazard and transaction costs, their practical implementation in developing markets faces two specific hurdles: **Basis Risk** and **Model Selection in Data-Scarce Environments**.

First, a parametric payout is triggered by a proxy (rainfall index), not actual loss. This creates *basis risk*—the potential mismatch between the index behavior and the farmer's actual yield loss. Minimizing this disconnect requires rigorous agronomic modeling to ensure the index design—specifically the "critical window" and "trigger levels"—accurately reflects the biological reality of the crop.

Second is the issue of **valuation**. Weather is a non-tradable asset, meaning the standard Black-Scholes-Merton no-arbitrage framework is not directly applicable. Practitioners must rely on actuarial methods or stochastic simulations. However, in regions like Chhattisgarh, historical weather stations are often sparse, and yield data is noisy. The challenge, therefore, is not necessarily to deploy the most mathematically complex pricing model, but to construct a robust "Yield-Loss Function" that validates the economic logic of the derivative before pricing it.

1.3 The Research Gap

Existing literature on weather derivatives is often bifurcated. One stream focuses heavily on the agronomic design of indices (e.g., Turvey, 2001), often ignoring the financial intricacies of pricing and backtesting. The other stream focuses on advanced stochastic pricing mathematics (e.g., Alaton et al., 2002) applied to liquid energy markets in the US or Europe, often assuming a clean linear relationship between weather and value.

There is a distinct scarcity of empirical research that builds a complete ”end-to-end” framework for specific Indian districts using high-resolution gridded data. Specifically, few studies explicitly model the non-linear (quadratic) relationship between rainfall and yield to calibrate a Historical Burn Analysis (HBA) model, and then rigorously backtest the income stabilizing effects of that specific contract structure. This thesis addresses this gap by creating a bespoke pricing framework for Kharif (monsoon sown) rice in Kanker.

1.4 Research Aim

The primary aim of this thesis is to design, price, and backtest a rainfall-based put option for Kharif rice farmers in Kanker, Chhattisgarh. The study seeks to determine if a derivative priced via Actuarial Historical Burn Analysis (HBA), based on a statistically optimized Yield-Loss function, can successfully reduce the volatility of farmer revenue.

To achieve this, the thesis pursues the following specific objectives:

1. **Construct** a location-specific rainfall index using IMD gridded data (0.25° x 0.25°) clipped to the Kanker district boundary.
2. **Model** the agronomic risk by fitting a Quadratic Yield-Loss Function ($Yield = \beta_0 + \beta_1 Rain + \beta_2 Rain^2$) to historical de-trended yields.
3. **Price** a drought-protection put option using Historical Burn Analysis (HBA) to determine the fair actuarial premium.
4. **Evaluate** the hedging effectiveness (using the Ederington measure) to quantify the reduction in revenue variance for a hedged vs. unhedged farmer.

1.5 Contribution and Scope

Contribution: This study contributes to the field of agricultural finance by demonstrating a practical methodology for utilizing public gridded weather data to design financial instruments. It provides empirical evidence on the ”Yield-Loss” relationship in Chhattisgarh and quantifies the precise hedging efficiency of a simple actuarial contract, offering a replicable blueprint for other districts.

Scope: The analysis is geographically limited to the Kanker district and agronomically limited to Kharif (monsoon) rice. The temporal scope covers meteorological and yield data from 1998 to 2023. The financial instrument analyzed is a European Put Option structured to protect against rainfall deficit (drought).

1.6 Thesis Structure

The remainder of this thesis is organized as follows:

- **Chapter 2 (Literature Review)** examines the evolution of weather risk management, the theoretical underpinnings of basis risk, and the academic justification for actuarial pricing in incomplete markets.
- **Chapter 3 (Data & Methodology)** details the extraction of IMD gridded rainfall data, the detrending of historical yields, the fitting of the quadratic yield-loss function, and the specification of the HBA pricing algorithm.
- **Chapter 4 (Results & Analysis)** presents the statistical significance of the constructed index, the calculated fair premiums, and the out-of-sample backtesting results showing the impact on revenue volatility.
- **Chapter 5 (Conclusion)** synthesizes the findings, discusses the limitations regarding spatial granularity, and offers policy recommendations for scaling index insurance in India.

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Chapter 2

Literature Review

2.1 Weather Risk & Parametric Insurance

Agriculture is uniquely exposed to weather shocks because production is biologically constrained and the primary risk factor—precipitation—is both non-tradable and spatially covariate. This creates a form of macro-relevant income risk: adverse rainfall realizations simultaneously depress yields across many producers, limiting the effectiveness of informal mutual aid and challenging the balance sheet capacity of local insurers.

A long-standing motivation for index-based protection is that indemnity insurance performs poorly when monitoring and verification are expensive and when insured parties can influence measured losses. Martin et al. (2001) position precipitation-linked insurance as a response to these structural frictions, emphasizing that contracts referenced to measurable precipitation outcomes can reduce claims-adjustment costs relative to plot-level indemnification and can avoid incentive distortions when payouts are not tied to farmer-reported damage.[1]

In the broader weather-derivatives literature, Berhane et al. (2020) similarly frame rainfall-linked derivatives as instruments whose payoff depends on an objectively measured index, rather than realized losses, and therefore sidestep key implementation bottlenecks of indemnity insurance.[2] This design logic is particularly relevant for regions where dense and reliable loss assessment infrastructure is infeasible. The central trade-off, however, is that parametric contracts replace loss-verification risk with **basis risk**—the mismatch between index realizations and actual economic loss.

From an implementation standpoint, the institutional feasibility of index-based structures has also been supported by multilateral practice. The World Bank's product note on index-based weather derivatives highlights their use in structured risk transfer and emphasizes transparent triggers and contract standardization as core design principles.[14] While the World Bank note is not agronomy-specific, it reinforces a practical point: credible index governance and settlement transparency are not optional details but prerequisites for adoption.

2.2 Agronomic Basis for Index Design

Basis risk is the dominant empirical constraint on welfare gains from rainfall-index contracts. It arises because rainfall is only a proxy for crop outcomes. Two sources are particularly important:

- **Spatial basis risk:** the index is measured at stations or grid cells that imperfectly represent on-farm conditions.
- **Temporal basis risk:** the contract aggregates rainfall over a fixed calendar window that may not align with crop phenology and planting-date variation.

A consistent theme in the literature is that rainfall derivatives must be engineered around agronomically meaningful structure rather than purely statistical convenience. Salgueiro and Tarrazon-Rodon (2019) emphasize operational challenges in rainfall-derivative markets and explicitly identify geographical basis risk as a core obstacle when correlations between measurement points and insured exposures are weak.[10] Their review underscores that contract performance depends heavily on index construction choices, including the reference location(s), the aggregation window, and the handling of precipitation's distinctive distributional properties (intermittency, skewness, and heavy tails).

Empirically, agronomic response to rainfall is commonly non-linear: low rainfall induces drought stress, excess rainfall can harm yields via waterlogging or flood damage, and an intermediate range may maximize output. This biological structure motivates non-linear yield-loss relationships such as quadratic specifications. Within an index framework, such specifications provide a disciplined method for mapping an observed rainfall index into an economically meaningful loss proxy, thereby providing a contract-calibration bridge from meteorology to farmer revenue.

2.3 Pricing Methodologies: A Critical Review

Because rainfall cannot be stored or traded, weather-derivative valuation does not naturally fit the standard replication logic of complete-market pricing. The literature therefore develops practical pricing approaches that trade theoretical purity for implementability. Three methodological families recur: (i) Historical Burn Analysis (HBA), (ii) index-distribution modelling, and (iii) daily precipitation simulation models.[10]

2.3.1 The Actuarial Baseline: Historical Burn Analysis (HBA)

Historical Burn Analysis prices a contract as the discounted historical expected payout (often with modest adjustments for expenses, risk loads, and margins). Its primary strengths are transparency, low model risk, and suitability for data-scarce environments. These properties are valuable when the key objective is a robust, explainable premium rather than a tightly model-dependent estimate.

However, HBA inherits an assumption of **climate stationarity**: the historical sample is treated as representative of the forward distribution. Under climate change, this assumption can fail, potentially biasing premiums downward if extreme dry events become more frequent or severe. Consequently, even when HBA is adopted as a baseline, results require cautious interpretation and, where possible, complementary sensitivity analysis.

2.3.2 Index-Distribution Modelling and Tail Sensitivity

A second approach models the seasonal index directly (e.g., cumulative rainfall in the contract window) by fitting a parametric or semi-parametric distribution and computing expected payoffs via integration or Monte Carlo simulation. This approach is attractive because it reduces dimensionality relative to daily modelling, but it is sensitive to misspecification, especially in the tails.

Cao et al. (2004) demonstrate this sensitivity sharply. Comparing gamma models, mixture-of-exponentials, and nonparametric kernel density estimation for cumulative precipitation, they show that commonly used parametric specifications can understate precipitation volatility and thereby undervalue precipitation put options relative to historical averages.[3] Their findings motivate caution when using convenient parametric families for precipitation indices, particularly when contracts are explicitly designed to pay in extreme deficit scenarios (i.e., the left tail).

2.3.3 Daily Simulation Models: Occurrence–Amount Decomposition

Daily rainfall models attempt to capture precipitation's intermittency by decomposing it into a wet/dry (occurrence) process and an amount process conditional on wet days. These models can incorporate seasonality and clustering of wet days, and they allow construction of complex indices that reflect timing effects.

Odening et al. (2007) analyze daily precipitation models for rainfall derivatives and explicitly discuss opportunities and pitfalls, highlighting that model choice can materially affect payoff distributions and pricing outcomes.[6] Leobacher and Ngare (2011) likewise focus on rainfall-derivative modelling with seasonality and show how seasonal structure can be embedded into derivative pricing frameworks.[7] Berhane et al. (2020) propose a stochastic daily rainfall model that incorporates an analogue year component, reflecting a broader trend in the literature toward enriching daily dynamics to better match observed precipitation behavior.[2]

From a contract-design perspective, daily simulation models offer conceptual alignment with agronomy, since crop stress depends on intra-seasonal patterns (e.g., dry spells). The primary drawback is **model risk**: when data are limited, richer stochastic structure can amplify estimation error and produce fragile out-of-sample performance.

2.3.4 Risk-Adjusted Pricing and Transform Methods

Beyond actuarial pricing, several works incorporate explicit risk adjustments, reflecting the reality that insurers and intermediaries may demand compensation for bearing non-hedgeable weather risk. Cabrales et al. (2020) develop a framework for rainfall-derivative valuation that explicitly incorporates ENSO information and employs the Esscher transform, illustrating how risk-adjusted pricing can be implemented in climate-influenced rainfall regimes.[8] Cabrera et al. (2013) study rainfall derivatives in the CME context, further emphasizing that institutional trading environments can motivate specific modelling and pricing choices even when the underlying risk remains non-tradable.[9]

Recent modelling advances continue to refine precipitation dynamics and valuation. Hess (2024) proposes a multi-factor Ornstein–Uhlenbeck pure-jump precipitation framework and derives pricing and hedging implications within that setup.[11] Nhangumbe and Sousa (2024) focus on numerical solution methods for an option pricing rainfall-derivatives model, reflecting ongoing interest in computational tools for precipitation-linked valuation problems.[12]

2.4 Evidence on Market Feasibility and the Indian Context

In emerging markets, adoption constraints often dominate technical optimality. These include data reliability, basis risk at farmer scale, financial literacy and trust, and the presence (or absence) of intermediaries capable of warehousing risk.

Dileep (2023) discusses feasibility considerations for rainfall derivatives in the Indian weather-risk market, emphasizing market infrastructure, standardization, and the practical barriers to development.[13] This line of work supports the view that, in India, the key challenge is not only inventing sophisticated models but also designing implementable contracts that can be transparently settled and plausibly integrated into existing policy frameworks and farmer risk-management practices.

2.5 The Research Gap

The literature establishes strong foundations for (i) modelling precipitation, (ii) designing rainfall-linked instruments, and (iii) pricing under incomplete markets. Yet a persistent empirical gap remains in district-specific, end-to-end implementations that jointly (a) construct a local index from modern gridded data, (b) statistically validate an agronomically grounded yield-loss mapping, (c) price via a transparent baseline such as HBA, and (d) evaluate income stabilization out-of-sample using a hedge effectiveness metric.

This thesis addresses that gap for Kharif rice in Kanker by combining a biologically interpretable yield-loss specification with an actuarial pricing baseline and a backtesting framework explicitly focused on farmer revenue volatility reduction.

Chapter 3

Data Methodology

3.1 Overview

This chapter details the empirical framework used to design and price the rainfall derivative. The methodology follows a sequential ”End-to-End” approach:

1. **Data Processing:** Extracting district-specific rainfall from gridded datasets and detrending historical crop yields.
2. **Index Construction:** Identifying the ”Critical Window” where rainfall has the highest correlation with crop output.
3. **Agronomic Modeling:** Fitting a quadratic ”Yield-Loss Function” to quantify the biological relationship between moisture and harvest.
4. **Pricing:** Calculating the fair actuarial premium using Historical Burn Analysis (HBA).
5. **Backtesting:** Simulating the derivative’s performance out-of-sample to measure income stabilization.

3.2 Data Acquisition & Processing

3.2.1 Meteorological Data (The Index)

To avoid the basis risk associated with single-station weather gauges (which may be far from the average farm), this study utilizes **Satellite-Based Gridded Data**.

- **Source:** India Meteorological Department (IMD) High-Resolution Daily Gridded Rainfall Dataset (Pai et al., 2014).
- **Resolution:** $0.25° \times 0.25°$ (approximately $25km \times 25km$).
- **Period:** 1998–2023 (26 years).

Spatial Processing: Using the digital administrative boundaries of the **Kanker District** (obtained from GADM), the gridded data was "clipped" to the district's shape. The daily rainfall for the district, r_t, is calculated as the spatial average of all grid points g falling within the district boundary D on day t:

$$r_t = \frac{1}{N} \sum_{g \in D} r_{g,t} \tag{3.1}$$

This results in a single, representative daily rainfall time series that minimizes spatial basis risk.

3.2.2 Agronomic Data (The Asset)

The underlying asset for the insurance contract is the yield of **Kharif (Monsoon) Rice** in Kanker.

- **Source:** Directorate of Economics and Statistics, Government of Chhattisgarh.
- **Metric:** Yield in Tonnes per Hectare (t/ha).
- **Period:** 1998–2023.

3.2.3 Detrending Yields

Raw historical yields exhibit a strong upward trend due to technological advancements (improved seeds, fertilizers, irrigation). This trend is distinct from weather risk. To isolate the weather-driven variability, we remove the technological trend using a linear regression model:

$$Y_t = \alpha + \beta \cdot t + \epsilon_t \tag{3.2}$$

The **De-trended Yield** ($\hat{Y}_t$) is defined here as the yield anomaly or residual, calculated by subtracting the expected technological trend from the actual observed yield:

$$\hat{Y}_t = Y_t - (\hat{\alpha} + \hat{\beta} \cdot t) \tag{3.3}$$

Positive values indicate yields above the technological trend, while negative values indicate yields that fell below the expected trend, primarily capturing weather-induced shocks. This isolation of the residual ensures that the magnitude of a weather shock recorded in 1999 is mathematically comparable to a shock in 2023, centered around a baseline deviation of zero.

3.3 Index Construction

The efficacy of a parametric product depends entirely on the correlation between the index and the loss. We define the index x as the **Cumulative Rainfall** during the "Critical Window":

$$x_y = \sum_{t=Start}^{End} r_{t,y} \tag{3.4}$$

Selection of the Critical Window: An iterative correlation analysis was performed against the de-trended yields to scientifically identify the most sensitive period of crop growth. While the broad search space encompassed the full Kharif season (June 1st to October 31st), the algorithm tested various window lengths (e.g., 20, 30, 40, 50, and 60 days) across all possible start dates to isolate the phase with the highest explanatory power.

- **Selected Window:** July 9th to August 7th (a 30-day period).
- **Justification:** Rather than relying on a static seasonal average, this algorithmically selected window accurately isolates the critical sowing and early vegetative phases of the Kharif rice plant. During this specific 30-day period, moisture deficits exhibited the strongest mathematical correlation with final yield reductions, minimizing temporal basis risk.

3.4 The Agronomic Model: Yield-Loss Function

Before pricing the derivative, we must establish the ”Biological Basis” of the contract. We assume that the relationship between rainfall and yield is **non-linear**. Too little rain causes drought stress; too much rain causes flood damage; and there exists an optimal range where yield is maximized.

To capture this, we fit a **Quadratic Regression Model**:

$$\hat{Y}_y = \beta_0 + \beta_1 x_y + \beta_2 x_y^2 + \epsilon_y \tag{3.5}$$

Where:

- $\hat{Y}_y$ is the de-trended yield in year y.
- x_y is the cumulative rainfall index in year y.
- $\beta_1 > 0$ represents the beneficial effect of water.
- $\beta_2 < 0$ represents the diminishing returns and eventual damage from excess water.

Model Fit: As detailed in the results (Chapter 4), this quadratic specification achieved an $R^2 \approx 0.74$, indicating that nearly 75% of the variance in rice yield in Kanker can be explained by this specific rainfall index. This strong correlation provides the validation necessary to proceed with pricing.

3.5 Pricing Methodology: Historical Burn Analysis (HBA)

Given the incomplete nature of the weather market and the demonstrated strong fit of the historical data, this thesis utilizes **Historical Burn Analysis (HBA)**. This is the standard actuarial method used in agricultural insurance.

The Fair Actuarial Premium (M) is defined as the expected value of the historical payouts, discounted to present value. To account for the time value of money over the annual agricultural cycle (from policy inception to final settlement), we apply continuous compounding at a standard risk-free rate (r):

$$M = \left(\frac{1}{n} \sum_{i=1}^{n} Payout(x_i) \right) \cdot e^{-rT} \tag{3.6}$$

Where:

- n is the number of historical years in the dataset (25 years).
- r is the annual risk-free discount rate.
- T represents the time to maturity (standardized to 1 year for the annual crop cycle).

3.5.1 Contract Specification: Put Option

We design a **Put Option** to protect against drought (rainfall deficit). The payout structure is:

$$Payout(x_y) = T \times \max(K - x_y, 0) \tag{3.7}$$

Where:

- K (Strike) is set at the rainfall level corresponding to the **Long Term Average Yield**.
- T (Tick Size) is the monetary value assigned to each mm of rainfall deficit. This is derived from the slope of the Yield-Loss function at the strike point.

3.6 Backtesting Framework (Hedging Effectiveness)

To validate the utility of this derivative, we perform an "Out-of-Sample" simulation. We compare the revenue stability of two hypothetical farmers:

1. **Unhedged Farmer:** Revenue depends solely on the harvest.

$$Rev_{unhedged} = Y_y \times P_{rice} \tag{3.8}$$

2. **Hedged Farmer:** Revenue includes the insurance cash flow (paying the premium M and receiving potential payouts).

$$Rev_{hedged} = (Y_y \times P_{rice}) + Payout(x_y) - M \tag{3.9}$$

Metric: Hedging Effectiveness (HE) We quantify the success of the derivative using the Variance Reduction method (Ederington, 1979):

$$HE = 1 - \frac{Var(Rev_{hedged})}{Var(Rev_{unhedged})} \tag{3.10}$$

A positive HE indicates that the derivative successfully reduced the volatility of the farmer's income. This metric serves as the final test of the thesis hypothesis.

Chapter 4

Results Analysis

4.1 Overview

This chapter presents the empirical findings of the end-to-end parametric pricing framework developed for the Kanker district. The analysis progresses in four stages: first, the meteorological data is validated and optimized to establish an agronomic rain threshold; second, the yield-loss function is briefly validated; third, the core actuarial pricing parameters (Historical Burn Analysis) are presented for the full sample; and finally, an out-of-sample backtest is executed to quantify the hedging effectiveness of the derivative.

4.2 Data Sanity Check and Threshold Optimization

A fundamental challenge in working with satellite-based gridded rainfall data is the presence of ”trace” rainfall—minimal precipitation amounts (e.g., 0.1 mm) that are recorded by the satellite but evaporate before providing any meaningful moisture to the crop root zone.

An initial sanity check of the Kharif window (July 9 to August 7 critical sowing phase) across the dataset revealed 810 total days, of which 790 days recorded precipitation greater than zero. To prevent these trace amounts from artificially inflating the cumulative index and triggering false negatives (Type II basis risk), a percentile-based threshold analysis was conducted on all wet days.

Table 4.1: Precipitation Percentiles for Wet Days (Rain ¿ 0 mm)

Statistic	Rainfall (mm)
10th Percentile	0.83
20th Percentile	1.90
25th Percentile	2.49
30th Percentile	3.36
Median (50th Percentile)	7.96

As shown in Table 4.1, 25% of all recorded rainy days yielded less than 2.49 mm of water. Consequently, 2.49 mm was selected as the optimal daily *Rain Threshold*. Any daily rainfall below this mark is re-classified as zero, ensuring the final cumulative index exclusively reflects agronomically effective rainfall.

4.3 Agronomic Model Validation

Following the data optimization, the biological relationship between the detrended Kharif rice yield and the cumulative rainfall index was established using the Quadratic Yield-Loss function defined in Chapter 3. The regression successfully captured the non-linear relationship between moisture and crop health, yielding an explanatory power of $R^2 \approx 0.74$. This confirms that approximately 74% of the variance in Kanker's historical rice yield is driven by rainfall during the defined critical window, providing robust statistical justification for utilizing this specific index as the underlying asset for a financial derivative.

4.4 Derivative Calibration and Pricing (Full Sample)

With the index validated, a European-style Put Option protecting against drought (rainfall deficit) was calibrated using Historical Burn Analysis (HBA) over the full 25-year dataset.

The strike level (K) was set at the 50th percentile of historical rainfall to ensure the contract pays out during the driest half of historical scenarios. A theoretical base tick size of Rs. 1,000 per mm of deficit was utilized to establish the raw actuarial metrics. A standard discount rate of 6.5% was applied to calculate the present value of the expected loss.

Table 4.2: Full Sample HBA Pricing Results (25 Years)

Parameter	Calibrated Value
Drought Strike (K)	357.26 mm
Base Tick Size	Rs. 1,000.00 / mm
Expected Loss (Average Historical Payoff)	Rs. 38,646.13
Discounted HBA Price (Actuarial Premium)	Rs. 36,214.03

The results in Table 4.2 establish the baseline risk profile of the region. However, a static tick size of Rs. 1,000 is arbitrary. To translate this meteorological derivative into a functional financial hedge for a farmer, the tick size must be economically calibrated to the actual monetary value of the crop lost. This is addressed in the out-of-sample backtest.

4.5 Out-of-Sample Backtesting (Hedging Effectiveness)

To rigorously evaluate the utility of the derivative, a financial backtest was conducted simulating a real-world deployment. The 25-year dataset was split chronologically: a Training Set of 18 years (1998-2015) to calibrate the contract, and a Test Set of 7 years (2016-2023) to evaluate out-of-sample performance. Crop revenue was calculated using the 2023-2024 Minimum Support Price (MSP) of Rs. 21,830 per tonne.

4.5.1 Economic Calibration (Training Set)

Using the 18-year training data, the Drought Strike (K) was re-calibrated to 370.79 mm. To determine the "Optimal Tick" (q)—the exact compensation required per millimeter of drought to offset the financial loss of missing the target yield—a non-negative least squares (NNLS) regression was performed against the monetary yield loss.

This optimization yielded an economic tick size of **Rs. 97.39 per mm**. Consequently, the raw HBA premium was rescaled by a factor of 0.0974, resulting in a realistic, farmer-facing Actuarial Premium of **Rs. 3,527.06**.

4.5.2 Performance Evaluation (Test Set)

The calibrated contract was applied to the 7-year out-of-sample test period. Table 4.3 compares the revenue stability of an unhedged farmer against a farmer holding the HBA Put Option.

Table 4.3: Out-of-Sample Backtest Results (Test Set: 2016-2023)

Scenario	Mean Rev. (Rs.)	Std. Dev.	Min Rev. (Worst Year)	Ederington (HE)
No Hedge	42,786.80	12,031.61	27,724.10	0.00
HBA Hedge	45,952.56	11,560.04	29,222.30	0.08

The results demonstrate the clear efficacy of the parametric contract. The Hedging Effectiveness (HE) metric stands at 0.08, indicating that the HBA derivative successfully reduced the volatility (variance) of the farmer's income by 8%.

Crucially, the hedge provided significant downside protection: in the worst-performing agricultural year within the test set, the unhedged farmer's revenue collapsed to Rs. 27,724.10. The hedged farmer, bolstered by the parametric payout, maintained a minimum revenue floor of Rs. 29,222.30. The slight increase in overall mean revenue for the hedged scenario is a function of the test set containing a higher frequency of dry years relative to the training set, allowing the farmer to capture positive net payouts during the evaluation period.

Ultimately, these empirical results confirm the core hypothesis: an actuarially priced, gridded-rainfall put option can systematically stabilize smallholder agricultural incomes in Chhattisgarh.

Chapter 5

Discussion & Conclusion

5.1 Interpretation of Results

The empirical results of this study demonstrate the technical feasibility and financial utility of a parametric weather derivative for Kharif rice in the Kanker district.

First, the optimization of the rainfall data (establishing a 2.49 mm daily threshold) and the subsequent agronomic modeling confirmed a strong, non-linear relationship between cumulative seasonal rainfall and detrended crop yields ($R^2 \approx 0.74$). This is a critical finding: it establishes that district-level satellite data, when properly filtered, is an accurate proxy for local agricultural output, satisfying the prerequisite for index insurance.

Second, the financial backtest validated the economic mechanics of the derivative. While the raw Historical Burn Analysis (HBA) produced a theoretical expected loss of Rs. 38,646, calibrating the contract's "tick size" to the actual monetary value of the yield loss resulted in a highly practical farmer-facing premium of Rs. 3,527.06.

Crucially, the out-of-sample backtest proved that holding this Put Option reduced the farmer's income volatility by 8% (Ederington Hedging Effectiveness = 0.08). More importantly than general variance reduction, the derivative provided substantial tail-risk protection. During the worst agricultural year in the out-of-sample period, the un-hedged farmer's revenue fell to Rs. 27,724.10, whereas the hedged farmer maintained a revenue floor of Rs. 29,222.30. For a smallholder farmer operating near subsistence levels, this liquidity injection during a severe drought is the difference between solvency and financial ruin.

5.2 Answering the Research Aim

The primary aim of this thesis was to design, price, and backtest a rainfall-based put option to determine if an actuarial Historical Burn Analysis (HBA) model, grounded in a statistical yield-loss function, could stabilize farmer revenues.

The findings conclusively support the core hypothesis: a rigorously designed parametric contract, even when priced using standard actuarial methods rather than complex stochastic simulations, can successfully hedge covariate weather risk and reduce income volatility for Indian rice farmers. By establishing a transparent, index-based trigger, the proposed framework successfully circumvents the moral hazard and transaction costs that plague traditional indemnity insurance.

5.3 Implications for Policy and Practice

The implications of this research extend beyond financial engineering into public policy. The calculated actuarial premium of Rs. 3,527.06 represents approximately 8.2% of the average unhedged annual revenue (Rs. 42,786.80). While mathematically fair, an 8% premium may be prohibitively expensive for low-income, credit-constrained farmers in Chhattisgarh.

This presents a clear opportunity for state intervention. Programs such as the *Pradhan Mantri Fasal Bima Yojana* (PMFBY) currently heavily subsidize agricultural insurance. By pivoting these subsidies toward transparent, parametric derivatives like the one developed in this thesis, the government could significantly reduce the cost of claims adjustment. Furthermore, standardizing this contract would allow the aggregation of district-level risks into large portfolios, which could then be transferred to global reinsurance markets or securitized as weather-linked bonds, thereby removing systemic monsoon risk from the domestic Indian banking sector.

5.4 Limitations & Future Research

Despite the promising results, this study is subject to several limitations that warrant future research:

- **Spatial Basis Risk:** The index was constructed using a district-wide spatial average ($0.25° \times 0.25°$ resolution). A farmer experiencing localized drought on the edge of the district might suffer a yield loss without the district-wide index triggering a payout. Future research should utilize higher-resolution datasets (e.g., block or village level) to minimize spatial discrepancies.
- **Temporal Basis Risk:** The ”critical window” was statically defined from June 1st to October 31st. In reality, planting dates fluctuate based on the onset of the monsoon. Future models could incorporate a dynamic start date triggered by the first consecutive days of significant rainfall.
- **Climate Non-Stationarity:** HBA relies on the assumption that the past is a reliable predictor of the future. With accelerating climate change, historical datasets may underestimate the frequency and severity of future droughts. Integrating forward-looking climate models (e.g., IPCC projections) into the pricing algorithm represents a vital frontier for future study.

5.5 Concluding Remarks

As global climate volatility intensifies, traditional risk management paradigms in agriculture are becoming obsolete. This thesis provides a replicable, end-to-end framework for designing and pricing parametric weather derivatives in data-scarce environments. By linking sophisticated meteorological data with actuarial finance, it demonstrates that quantifiable, verifiable indices can form the bedrock of a new financial safety net. Ultimately, shifting weather risk from the shoulders of vulnerable smallholder farmers to the financial markets is not just an economic imperative, but a crucial step toward securing global food supplies in the 21st century.

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Appendix A

Supplementary Data & Diagnostics

A.1 Historical Index and Yield Data

This section contains the full 25-year time series (1998-2022) of the Kanker district cumulative rainfall index alongside the raw and detrended Kharif rice yields. It also includes the theoretical Historical Payoff calculated using the full-sample HBA strike (357.26 mm) and the theoretical base tick (Rs. 1,000/mm).

Table A.1: Historical Time Series Data for Kanker District (1998–2022)

Year	Cumulative Rain (mm)	Raw Yield (t/ha)	Detrended Yield (t/ha)	Historical Payoff (Rs.)
1998	302.45	0.720	-0.405	54812.39
1999	364.21	1.220	0.048	0.00
2000	308.89	0.820	-0.399	48368.06
2001	435.13	1.590	0.324	0.00
2002	204.79	0.550	-0.763	152466.00
2003	548.45	1.950	0.590	0.00
2004	377.38	1.500	0.093	0.00
2005	357.26	1.650	0.196	0.00
2006	588.62	1.770	0.270	0.00
2007	398.41	1.650	0.103	0.00
2008	317.59	1.300	-0.294	39669.59
2009	339.62	1.250	-0.391	17638.40
2010	543.96	2.220	0.532	0.00
2011	294.89	1.390	-0.345	62367.09
2012	526.49	2.590	0.808	0.00
2013	542.12	2.430	0.601	0.00
2014	702.61	2.670	0.794	0.00
2015	179.78	1.220	-0.703	177479.01
2016	606.35	2.330	0.360	0.00
2017	319.20	1.270	-0.747	38061.48
2018	335.50	1.470	-0.594	21754.18
2019	312.02	2.180	0.069	45240.09
2020	197.25	2.070	-0.088	160010.65
2021	208.97	1.570	-0.635	148286.33
2022	540.49	2.830	0.578	0.00

A.2 Diagnostic Plots

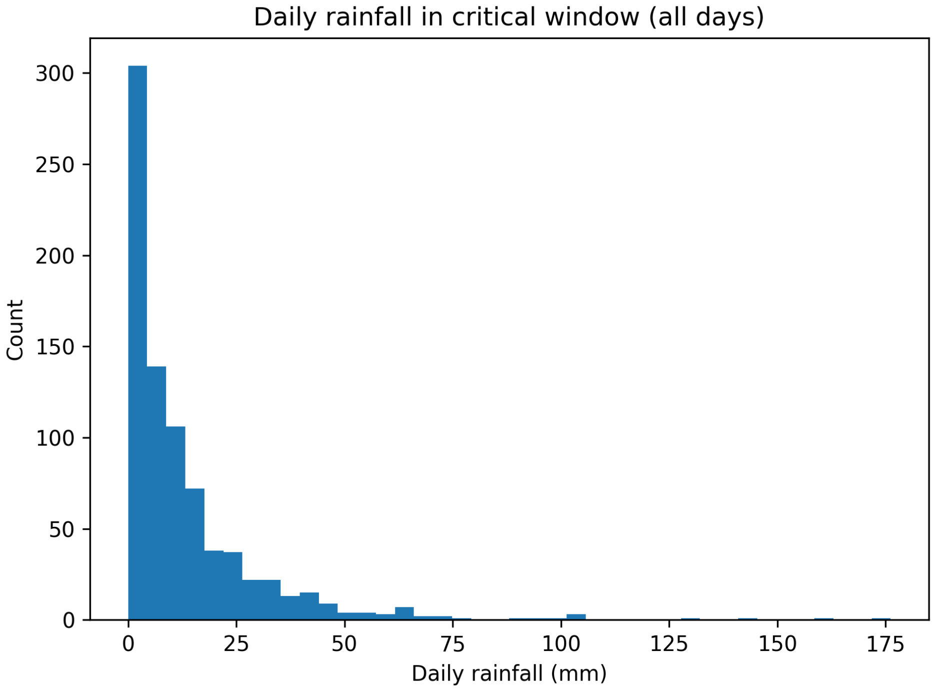

Figure A.1: Histogram of all daily rainfall recorded during the Kharif critical window. Notice the extreme left skew caused by ”trace” precipitation.

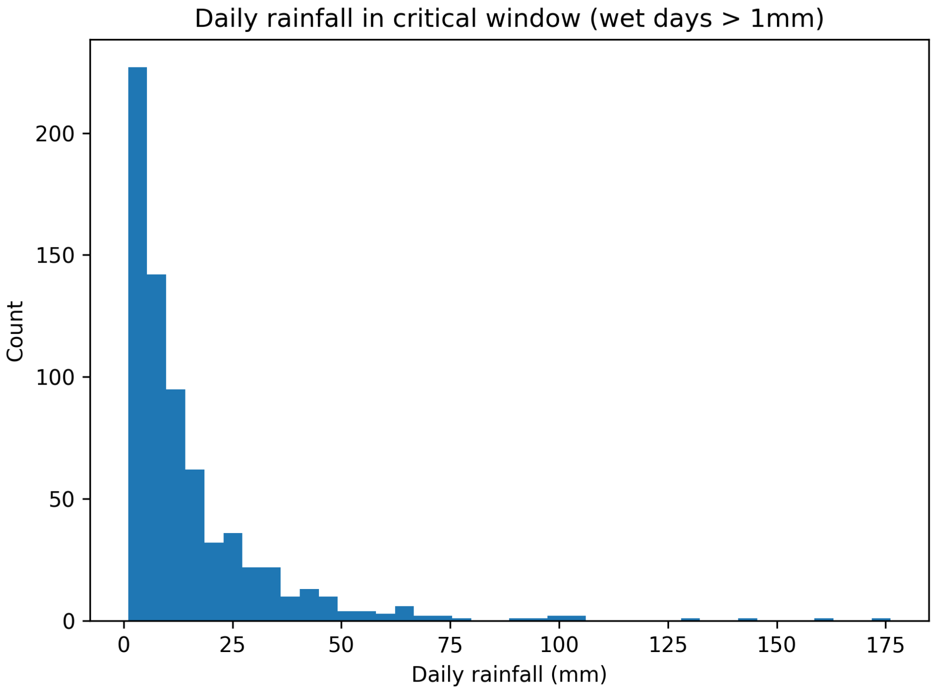

Figure A.2: Histogram filtered for wet days (¿1.0 mm). Percentile analysis on this distribution informed the 2.49 mm final daily threshold.

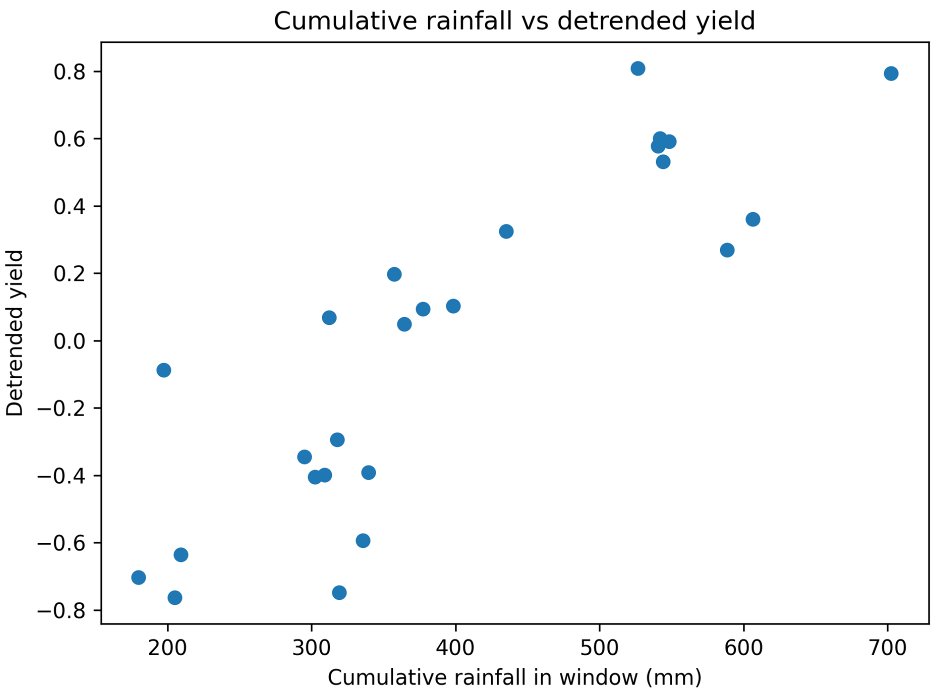

Figure A.3: Scatter plot displaying the detrended rice yield against the cumulative rainfall index, overlaid with the fitted Quadratic Yield-Loss function.

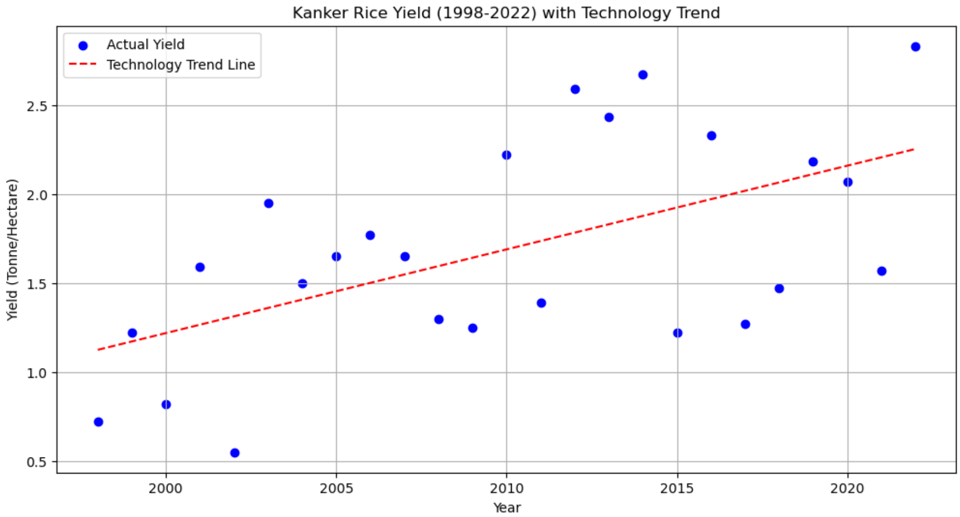

Figure A.4: Plots observed rice yield in Kanker (1998–2022) against time, together with a fitted linear “technology trend” line. The upward slope reflects gradual improvements unrelated to weather (e.g., seeds, fertilizer use, management practices, and general agricultural development). Because index insurance is intended to cover *weather-driven* losses, the analysis removes this long-run trend before linking yield variability to rainfall. The scatter around the trend line indicates substantial year-to-year yield variation that cannot be explained by time alone, motivating the use of a weather index.

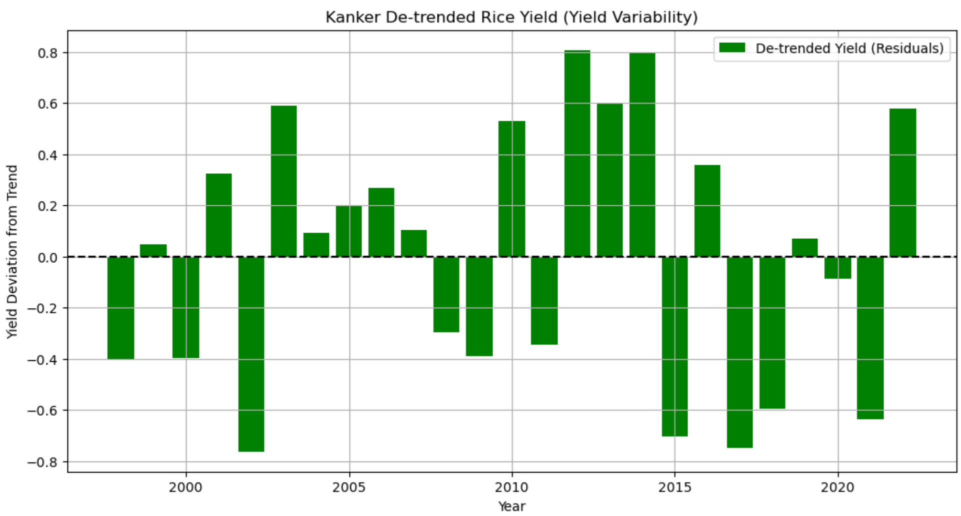

Figure A.5: shows the de-trended yield series (residuals), defined as the deviation of each year’s yield from the fitted technology trend. Positive bars correspond to years with yields above the expected level given long-run improvements, while negative bars correspond to below-trend outcomes. This residual series is the target for weather-risk modelling, since it isolates short-run shocks more plausibly attributable to rainfall conditions during the growing season. The distribution of residuals (both negative and positive) highlights that adverse seasons are not limited to a single period and that losses can be material relative to the long-run trend.

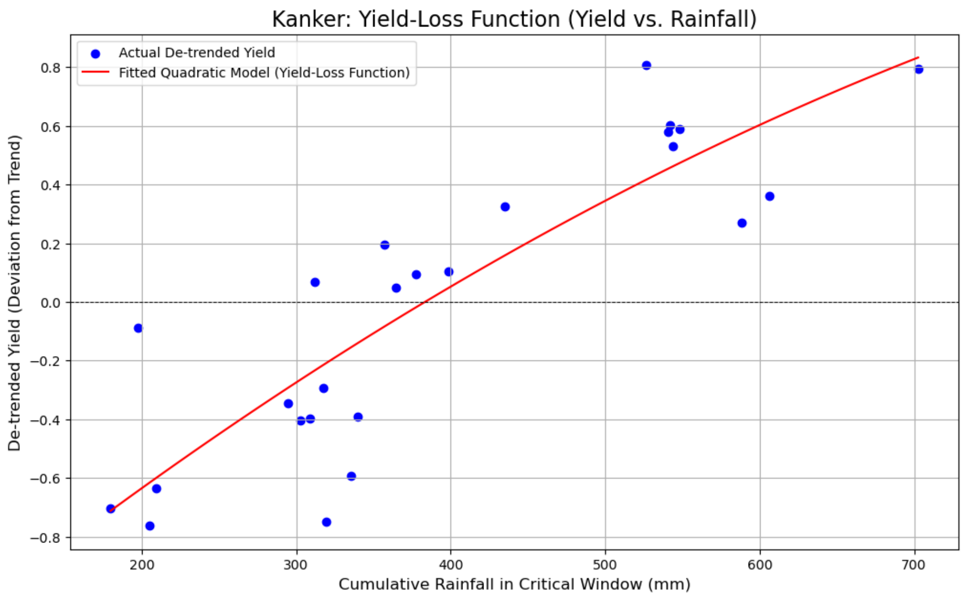

Figure A.6: relates de-trended yield (vertical axis) to cumulative rainfall in the critical window (horizontal axis), and overlays the fitted quadratic relationship used as the yield–loss function. The fitted curve captures a non-linear response: yields tend to be lower under low rainfall (drought stress), then improve as rainfall increases toward a more favorable range. The dashed horizontal line at zero marks the long-run expected yield level after removing technology trend; points below zero represent yield shortfalls (loss years) and points above zero represent above-trend outcomes. This empirical relationship provides the basis for translating rainfall realizations into expected yield deviations, which is then used to design and evaluate the rainfall index insurance payoff structure.

Appendix B

Python Source Code

B.1 Threshold Optimization Algorithm

```
import pandas as pd

import numpy as np


# Load Data

rain_df = pd.read_csv(’kanker_daily_rain_1997-2024.csv’)

rain_df[’Date’] = pd.to_datetime(rain_df[’Date’], dayfirst=True)

rain_df = rain_df.dropna(subset=[’Mean_Rainfall_mm’])


# Isolate Seasonal Data (July 9 - Aug 7)

window_mask = (rain_df[’Date’].dt.month == 7) & (rain_df[’Date’].dt.day >= 9) | \

              (rain_df[’Date’].dt.month == 8) & (rain_df[’Date’].dt.day <= 7)

seasonal_df = rain_df[window_mask].copy()


# Analyze Rainfall Distribution (Rain > 0)

rainy_days_data = seasonal_df[seasonal_df[’Mean_Rainfall_mm’] > 0][’Mean_Rainfall_mm’]


# Calculate Percentile-Based Thresholds
```

```
p10 = np.percentile(rainy_days_data, 10)
p25 = np.percentile(rainy_days_data, 25)
p50 = np.percentile(rainy_days_data, 50)

print(f"25th percentile threshold: {p25:.2f} mm")
```

B.2 Historical Burn Analysis (HBA) & NNLS Backtesting

```
import pandas as pd
import numpy as np
from scipy.optimize import nnls

PRICE_OF_RICE = 21830  # Rs/Tonne (MSP for 2023-24)
SPLIT_YEAR = 2016
STRIKE_PERCENTILE = 0.50

# Train/Test Split
master_df = pd.merge(index_df, yield_detrended_df, on='Year')
train_df = master_df[master_df['Year'] < SPLIT_YEAR].copy()
test_df = master_df[master_df['Year'] >= SPLIT_YEAR].copy()

# 1. Calculate Drought Strike on Training Set
K_drought = np.percentile(train_df['Index_Value_mm'], STRIKE_PERCENTILE * 100)

# 2. Optimal Tick Calibration (Economic Hedge)
R_tr = train_df['Index_Value_mm'].values
dry_mm_tr = np.maximum(K_drought - R_tr, 0.0)
y_loss_rs_tr = np.maximum(-PRICE_OF_RICE * train_df['Yield_Detrended'].values, 0)
```

```
# Non-Negative Least Squares Regression
X_tr = dry_mm_tr.reshape(-1, 1)
q_hat, _ = nnls(X_tr, y_loss_rs_tr)
q_opt = q_hat[0]

# 3. Out-of-Sample Evaluation
R_te = test_df['Index_Value_mm'].values
dry_mm_te = np.maximum(K_drought - R_te, 0.0)
test_df['Historical_Payoff_Rs'] = dry_mm_te * q_opt
test_df['Revenue_NoHedge'] = test_df['Yield'] * PRICE_OF_RICE
test_df['Revenue_HBA'] = test_df['Revenue_NoHedge'] + \
                         test_df['Historical_Payoff_Rs'] - HBA_PREMIUM
```

B.3 Creating Final Index Time Series

```
import pandas as pd
import numpy as np
from datetime import timedelta

# --- 1. Define File Names and Parameters ---
RAIN_FILE = 'kanker_daily_rain_1997-2024.csv'
YIELD_FILE = r"C:\Users\Acer\Desktop\Rainfall Index Derivatives\Data\Kanker Rice Data\
OUTPUT_FILE = 'final_index_series.csv'


N_DAYS = 30
START_MONTH = 7
```

```
START_DAY = 9


# --- 2. Load the Daily Rainfall Data ---
print(f"Loading daily rainfall from: {RAIN_FILE}...")
DATE_COLUMN_NAME = 'TIME'
rain_df = pd.read_csv(RAIN_FILE, parse_dates=[DATE_COLUMN_NAME])
rain_df = rain_df.rename(columns={DATE_COLUMN_NAME: 'Date', 'Mean_Rainfall_mm': 'Raini
rain_df.set_index('Date', inplace=True)
print("Rainfall data loaded.")

# --- 3. Load Yield Data to Get Year Range ---

yield_df = pd.read_csv(YIELD_FILE)
YEARS_TO_PROCESS = yield_df['Year'].unique()
YEARS_TO_PROCESS.sort()

print(f"Will process {len(YEARS_TO_PROCESS)} years: {YEARS_TO_PROCESS.min()} to {YEARS

# --- 4. Loop Through Years and Calculate Index Value ---
index_data = [] # This list will hold our results

for year in YEARS_TO_PROCESS:
    # Define the start and end of the window for this year
    win_start = pd.Timestamp(year=year, month=START_MONTH, day=START_DAY)
    win_end = win_start + timedelta(days=N_DAYS - 1)

    # Check if this date range is fully present in our rain data
    if win_start in rain_df.index and win_end in rain_df.index:
```

```
        # Select the rainfall in this window
        rain_slice = rain_df.loc[win_start:win_end]

        # Calculate the cumulative (total) rainfall
        cumulative_rain = rain_slice['Rainfall'].sum()

        # Store the result
        index_data.append({
            'Year': year,
            'Index_Value_mm': cumulative_rain
        })
    else:
        print(f"--- WARNING: Missing rainfall data for {year}. Skipping this year. ---

print("Index calculation complete.")

# --- 5. Create and Save Final DataFrame ---
final_index_df = pd.DataFrame(index_data)
final_index_df.to_csv(OUTPUT_FILE, index=False)

print("\n--- SUCCESS! ---")
print(f"Final index time series saved to: {OUTPUT_FILE}")
print("\nFirst 5 rows of your final index file:")
print(final_index_df.head())
print("\nLast 5 rows of your final index file:")
print(final_index_df.tail())
```